



AI reinvention made real

Accenture Innovation Challenge 2026

Team details

TEAM NAME: Digital Ninja



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College: IIT Kanpur
Stream: Mathematics and Scientific Computing
Year of graduation: 2027



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Describe the problem statement

Triage is a one-time snapshot taken on a patient who is still changing.

An emergency department rarely lacks equipment - it lacks sequenced attention. A nurse must rank strangers in minutes, and that ranking is then frozen while patients keep deteriorating in the waiting room.

The evidence is unambiguous. Across 5.3 million adult encounters, triage using the Emergency Severity Index was wrong in 32.2% of cases and caught only 65.9% of patients who went on to need a life-stabilising intervention. Undertriage is also unequal: Black patients carried an 18.5% higher relative risk of it.

Delay then converts error into death. Across 5.2 million NHS admissions, mortality rose steadily beyond five hours - one extra death for every 82 patients held 6-8 hours. In septic shock, each hour of delay added 1.8% absolute mortality.

The hardest patients are the quiet ones: 12.7% of confirmed heart attacks arrive without chest pain - 16.2% of women - and they die at over twice the rate. So the design question is not "**who is sickest now?**" but "**who is becoming sickest while nobody is looking?**"

THE EVIDENCE

32.2%

of 5,315,176 adult ED encounters were mistriaged under ESI — 3.3% undertriaged, 28.9% overtriaged. Sensitivity for high-acuity illness: **65.9%**.

Sax et al., JAMA Network Open, 2023 (Kaiser Permanente N. California, 21 EDs, 2016–2020)

1 in 82

patients delayed 6-8 hours from arrival dies as a result. 30 day mortality rises linearly past **5 hours** — a dose-response effect.

Jones et al., Emergency Medicine Journal, 2022 (5,249,891 patients, all major EDs in England)

+1.8% / hour

absolute mortality added per hour of delayed antibiotics in septic shock (adjusted OR **1.09** per elapsed hour across all sepsis).

Liu et al., Am. J. Respiratory & Critical Care Medicine, 2017 (35,000 sepsis patients)

12.7%

of confirmed heart attacks present **without chest pain** (16.2% of women vs 11.0% of men). 5-year mortality hazard ratio **2.12**.

Björck et al., Open Heart, 2018 (SWEDHEART registry, 172,981 patients)

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Proposed solution

ATRIA — a live queue, not a label. It may escalate on its own. It may never de-escalate on its own.

ATRIA sits beside the triage nurse and re-ranks every waiting patient continuously from vitals already being recorded. It changes **order of attention**, not clinical treatment.

Layer 0 : Red-flag gate (deterministic). 10 hard rules bypass the queue instantly: GCS ≤ 8 , SpO2 < 90%, SBP < 90 mmHg, active seizure, airway compromise, uncontrolled haemorrhage, anaphylaxis, stroke signs in window, eclampsia, paediatric retractions. No model can suppress a red flag.

Layer 1 : Acuity scorer (recommends). ~12 fields available in five minutes - arrival mode, complaint, HR, BP, SpO2, RR, temp, GCS, glucose, age, comorbidities - plus vulnerability weights for age < 15 and > 60.

Layer 2 : Dynamic re-ranker. Continuous SpO2 / BP / pulse, plus queue-aging so a long wait cannot be ignored, and re-assessment timers that force a second look.

Layer 3 : Human authority. The nurse signs off; every downgrade is hers alone.

Success metric: critical undertriage rate - audited per age, sex and language subgroup, because a widely used US risk algorithm was found to reduce Black patients identified for extra care from 46.5% to 17.7% by using cost as a proxy for need.

Obermeyer et al., Science, 2019. Human-oversight requirement per WHO *Ethics & Governance of AI for Health*, 2021.

DESIGNED FOR THE WORST CASE

Unknown $\neq$ Normal. A missing vital is imputed to the worst plausible value, never the median - median imputation is a silent undertriage engine.

Asymmetric loss. A false positive costs minutes. A false negative costs a life. We optimise for near-zero critical undertriage and accept overtriage as its purchase price.

Degraded mode. On outage, Layer 0 alone runs offline - deterministic, no ML.

DECIDE VS. RECOMMEND

System decides	Human decides
Firing red-flag alerts	Final acuity sign-off
Ordering within one band	Any downgrade of a flag
Re-assessment timers	Disposition & discharge
Flagging missing data	Triage under true scarcity



1. Continuous Ingestion

ATRIA ingests live HL7 ADT (Admissions) and ORU (Observational) feeds. As soon as a vitals machine captures data or a triage nurse types a chief complaint, it is securely encrypted and streamed to the ATRIA processing layer.



2. Real-Time Engine

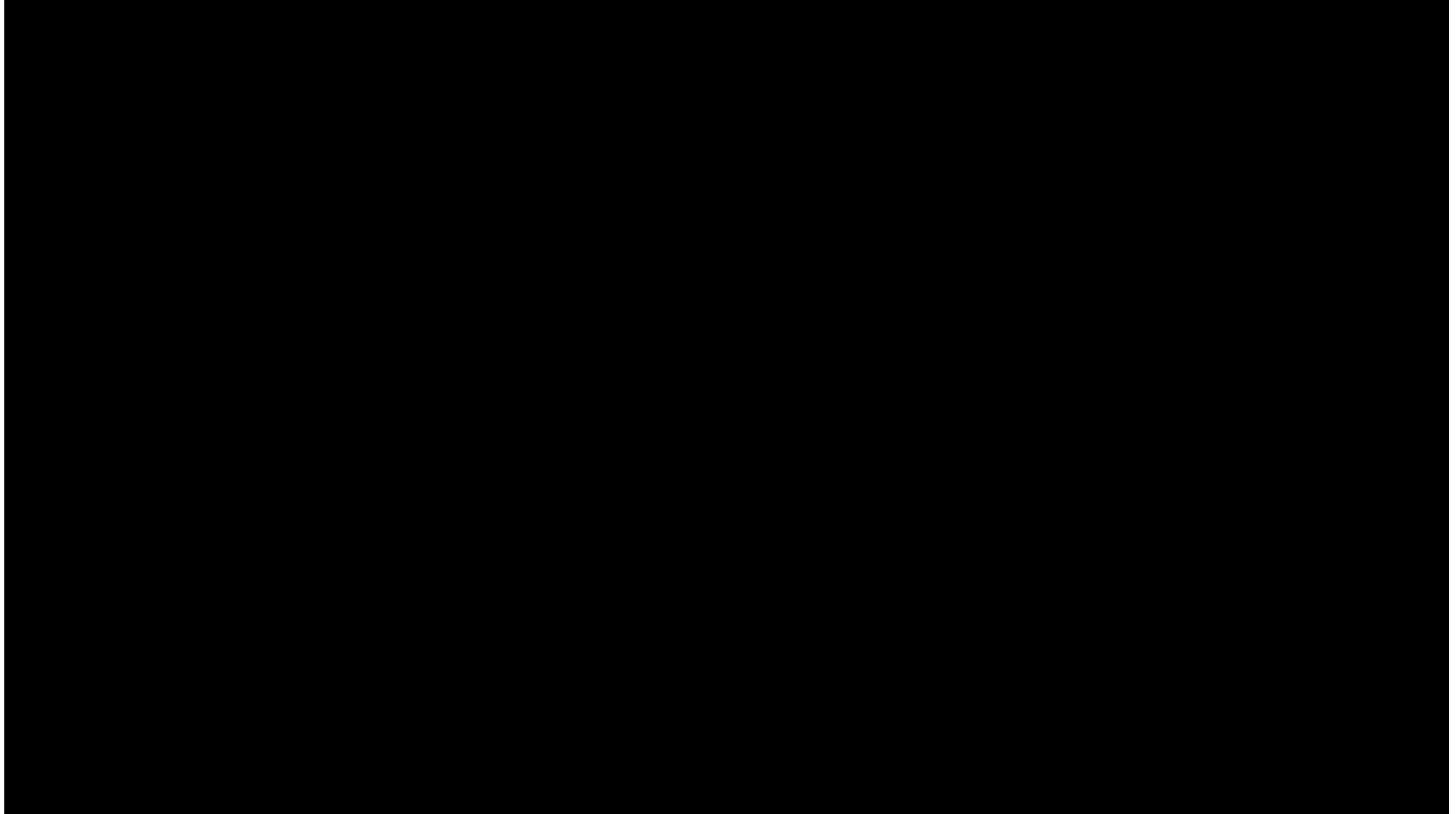
The models apply Layer 0 hard-rules and Layer 1/2 risk scoring simultaneously. Patient trajectories are mapped against historical baselines to identify subtle deteriorations in under 400 milliseconds.



3. Native Writeback

Escalations and re-rankings are pushed directly back into the EHR's native ED Track Board (e.g., Epic ASAP). The nurse sees a visual priority indicator in the software they already use every day.

Video



Thank you